Why the orbital phases allow—or block—the reaction

Reactant frontier orbitals → proposed transition state → product connectivity

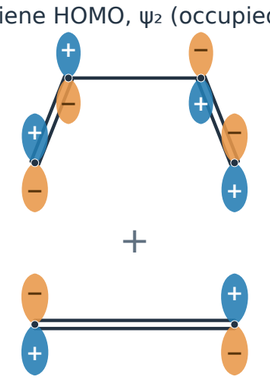
+ / − orbital phase (not charge) · forming overlap / bond · ✕ phase mismatch

Diels–Alder [4+2]

THERMALLY ALLOWED

suprafacial / suprafacial

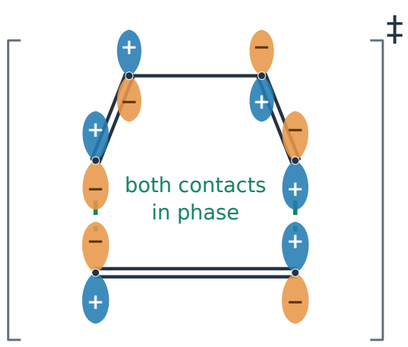
Diene HOMO, ψ₂ (occupied)



Ethene LUMO, ψ₂ (empty)

Δ

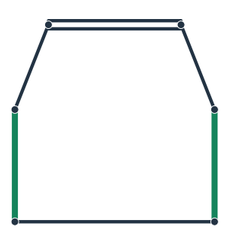
Hückel circuit · 6 electrons



Partial bonds join the facing lobes.

Δ

Product connectivity



Cyclohexene

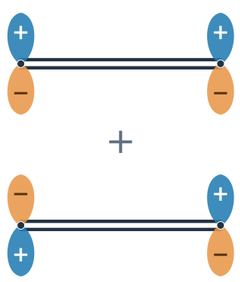
New σ bond(s) highlighted in green.

[2+2] cycloaddition

THERMALLY FORBIDDEN

suprafacial / suprafacial

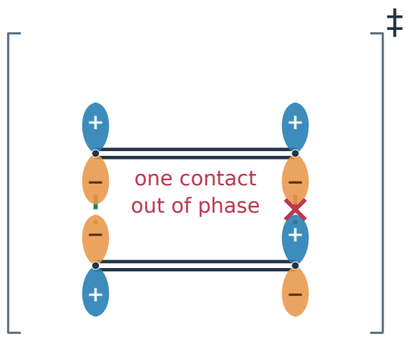
Ethene HOMO, ψ₁ (occupied)



Ethene LUMO, ψ₂ (empty)

Δ

Hückel circuit · 4 electrons



Partial bonds join the facing lobes.

Proposed product connectivity

Cyclobutane

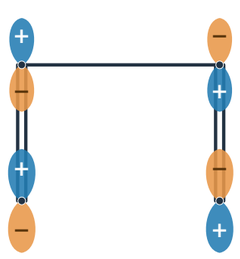
Forbidden for this concerted thermal route.

4π electrocyclic closure

THERMALLY ALLOWED

conrotatory

Polyene HOMO, ψ₂ (occupied)

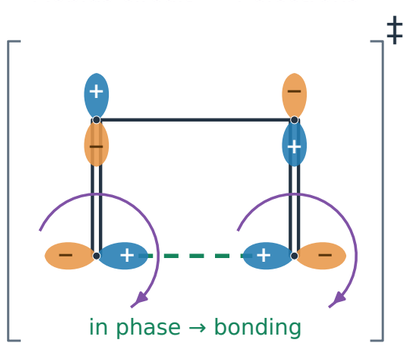


Opposite terminal phases

Follow the terminal lobes during rotation.

Δ

Möbius circuit · 4 electrons



same rotation sense

Product connectivity

Cyclobutene

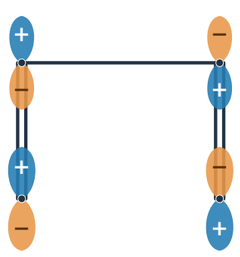
New σ bond(s) highlighted in green.

4π electrocyclic closure

THERMALLY FORBIDDEN

disrotatory

Polyene HOMO, ψ₂ (occupied)

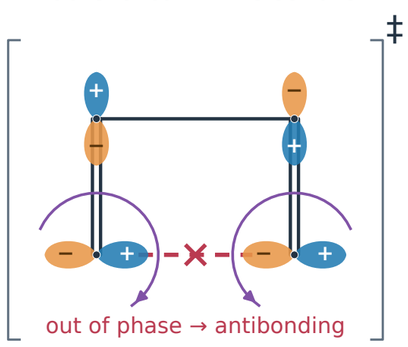


Opposite terminal phases

Follow the terminal lobes during rotation.

Δ

Hückel circuit · 4 electrons



opposite rotation senses

Proposed product connectivity

Cyclobutene

Forbidden for this concerted thermal route.

Hückel + 6 electrons and Möbius + 4 electrons are thermally allowed. Red pathways are forbidden for the specified concerted approach.

Schematic orbital phases, not calculated transition-state densities. A crossed arrow does not prohibit a different mechanism or photochemical route.